- X-Ray diffraction analysis is the most powerful tool for the determination of crystal structure. The inter planar distance (d) between two successive planes of atoms can be calculated using the following equation form the X-Ray diffraction data $2d\sin\theta = n\lambda$
- The structure of an ionic compound depends upon the stoichiometry and the size of the ions.generally in ionic crystals the bigger anions are present in the close packed arrangements and the cations occupy the voids. The ratio of radius of cation and anion $\left(\frac{r_{C^+}}{r_{A^-}}\right)$ plays an important role in determining the structure
- Crystals always found to have some defects in the arrangement of their constituent particles.
- Schottky defect arises due to the missing of equal number of cations and anions from the crystal lattice.
- Frenkel defect arises due to the dislocation of ions from its crystal lattice. The ion which is missing from the lattice point occupies an interstitial position.
- Metal excess defect arises due to the presence of more number of metal ions as compared to anions.
- Metal deficiency defect arises due to the presence of less number of cations than the anions.

EVALUATION

JN28IA

Choose the best answer:

1. Graphite and diamond are

   a) Covalent and molecular crystals b) ionic and covalent crystals

   c) both covalent crystals d) both molecular crystals

2. An ionic compound A_xB_y crystallizes in fcc type crystal structure with B ions at the centre of each face and A ion occupying corners of the cube. the correct formula of AxBy is

   a) AB b) AB_3

   c) A_3B d) A_8B_6

3. The ratio of close packed atoms to tetrahedral hole in cubic packing is

   a) 1:1 b) 1:2

   c) 2:1 d) 1:4

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